

Prajna G. Malettira



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OBJECTIVE

PhD researcher in Electrical and Computer Engineering at Purdue University specializing in model pruning, Neural Architecture Search (NAS), and PyTorch-based model compression. Proven track record of developing hardware-aware pruning and retraining strategies that accelerate inference speed and reduce memory footprints across diverse architectures, from large foundational models to vision networks. Seeking to leverage my deep background in model optimization and vision metrics to enhance edge and endpoint AI workloads.

EDUCATION

Purdue University, West Lafayette, Indiana, USA

Aug., 2022 - Present

Ph.D., Electrical and Computer Engineering

- **Focus:** Model Compression, Neural Architecture Search and Mechanistic Interpretability
- **Advisor:** Dr. Kaushik Roy · SRC Research Scholars Program

PES University, Bengaluru, Karnataka, India

Aug., 2018 - May, 2022

B.Tech., Electronics and Communication Engineering

SKILLS

Programming & ML: Python · Bash · PyTorch · HuggingFace Transformers & Diffusers · NumPy · scikit-learn

Distributed & Large-Scale: PyTorch DDP · DeepSpeed · TensorBoard

Platforms & Tools: Linux · SLURM GPU clusters · Git/GitHub · Conda

Vision Metrics & Tools: Top-1/Top-5 Accuracy · FID · CLIP Score · Torchvision · TensorBoard

PUBLICATIONS

- **Malettira, P. G., Nagaraj, M., Roy, A., Negi, S., & Roy, K.** "TraceNAS: Zero-shot LLM Pruning via Gradient Trace Correlation." **Under Review at NeurIPS 2026.** [\[Paper\]](#)
- **Malettira, P. G., Negi, S., Ponghiran, W., & Roy, K.** "TSkips: Efficiency through explicit temporal delay connections in spiking neural networks." **Accepted at TMLR 2025.** [\[Paper\]](#)
- **Roy, A., Malettira, P. G., Nagaraj, M., & Roy, K.** (2026). "TopoPrune: Robust Data Pruning via Unified Latent Space Topology." **Under Review at NeurIPS 2026.** [\[Paper\]](#)
- **Roy, A., Malettira, P.G., Nagaraj, M. and Roy, K.,** 2025. Unifying Global Topology Manifolds and Local Persistent Homology for Data Pruning. **Accepted at NeurIPS 2025 Symmetry and Geometry in Neural Representations Workshop.** [\[Paper\]](#)

RESEARCH EXPERIENCE

Geometric Diffusion Pruning (TGIP): Curvature-Aware Compression of Generative Models 2026-Present

Python, Multimodal Diffusion Transformers, Flow Matching, Differential Geometry, Graph Neural Networks WIP

- Developed a **training-free pruning framework for multimodal diffusion transformers** using local spectral isometric correction, enabling interpretable analysis of weight retention and bridging the gap between model compression and structural auditing.
- **Key takeaway:** Demonstrated that the method consistently outperformed SOTA baselines on Stable Diffusion models, **achieving higher CLIP and lower FID scores with 30–60% pruned models** in unstructured, structured and semi-structured settings.

Exploring Model Memorization and Data Leakage via Sparse Autoencoders 2026-Present

Python, SAEs, Mechanistic Interpretability, Dictionary Learning, Influence Functions WIP

- Designed an interpretability-driven framework with **sparse autoencoders and influence functions** to probe how individual training samples are encoded and memorized in vision models (ResNet & ViT), and **identified key concepts that drive memorization.**

- Developed a **lightweight memorization proxy** that operates without full model retraining, enabling learned representation interpretation for CIFAR-100 and ImageNet tasks.
- **Key takeaway:** Reported preliminary findings showing a **0.66 similarity with established memorization baselines**. Outlined next steps for methodology applications, SAE concept-level attribution, and extending the approach to language models.

TraceNAS: Zero-shot LLM Pruning via Gradient Trace Correlation

2025-2026

Python, Zero-shot NAS, LLM Pruning, Model Compression, Pretraining, Finetuning

Under Review

- Developed a **scalable, training-free NAS framework for sparse pruned LLMs**, enabling discovery of high-performing non-uniform architectures across **LLaMA, Qwen, and Mixture of Experts (MoE) models** at 2B to 70B parameter scales.
- Proposed a statistically grounded, scale-invariant, training-free proxy for pruned model performance, achieving high correlations of ($\rho = 0.94$) with downstream accuracy, perplexity and knowledge retention achieved after retraining with knowledge distillation.
- **Key takeaway:** Achieved a **4.22× prefill speedup, up to 2.5× lower peak memory usage, and 10× faster architecture discovery**, demonstrating empirical methodology at large model scale

TopoPrune: Robust Data Pruning via Unified Latent Space Topology

2025-2026

Python, Topological Data Analysis, Persistent Homology

Under Review; [NeurIPS 2025 Workshop](#)

- Developed a **novel framework for topologically grounded dataset pruning**, resolving instability in traditional geometric data pruning methods and producing highly representative data subsets.
- Validated compression strategy across standard vision backbones (ResNet, ViT, and OpenCLIP) on benchmark datasets including CIFAR-10/100 and ImageNet.
- **Key takeaway:** Improved coreset vision model accuracy by up to **1.5%** and precision by up to **6.5×** at high pruning rates, tracking standard **vision accuracy metrics (Top-1/Top-5)** to guarantee robust downstream performance.

TSkips: Efficiency Through Explicit Temporal Delay Connections in Spiking Neural Networks

2023-2024

Python, Neuromorphic Computing, Zero-shot NAS, Temporal Augmentation

[TMLR 2025](#)

- Developed a novel training mechanism for Spiking Neural Networks (SNNs) capturing long-term spatio-temporal patterns; achieved **56% reduced inference energy** in optical flow estimation and up to **16%** accuracy improvement in gesture and speech recognition.
- Leveraged Neural Architecture Search (NAS) with zero-shot performance estimation for memory-efficient exploration of temporal delay configurations.
- **Key takeaway:** Demonstrated **substantial performance gains in temporal vision tasks** directly applicable to autonomous navigation, achieving a **10% reduction in average end-point error for optical flow estimation** and up to a **16.04%** accuracy improvement in sequential data processing.

TEACHING & MENTORING

Teaching Assistant, EPICS Program, Purdue University

Aug., 2023-May, 2024

- Mentored **4 student groups (20–25 students each)** on open-ended engineering projects addressing sustainability challenges in the Lafayette community, providing technical guidance and managing project milestones under significant ambiguity.

HONORS & AWARDS

- **Millennium Fellow**, Millennium Campus Network & UN Academic Impact 2021
Selected to further UN Sustainable Development Goals via project *Anzen: Your Friend in the Dark*.
- **MRD Merit Scholarship**, PES University 2021 & 2022
Awarded for ranking in the top 20% of class.

RELEVANT COURSEWORK

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|-------------------------|--------------------|-----------------------|---------------------|
| • Computational Imaging | • Computer Vision | • Deep Learning (DL) | • Generative Models |
| • Optimization for DL | • Random Variables | • Pattern Recognition | |